CYPRESS
EMBEDDED SYSTEMS

PowerMonitor

AC Line Voltage Monitoring Device

User Manual

10002-UM-001

Revision 1

Table of Contents

Revision History 3
Approvals 3
Introduction 4
Welcome 4
Warranty Coverage 4
Battery Liability Disclaimer 5
Cellular Connectivity and Return Policy 5
General Considerations and Safety Information 6
Document Notation 6
Document Conventions 6
Operating Environment 6
Assumptions 6
Cleaning 6
Lithium Batteries 8
Periodic Checks 9
Device Features 10
External Features 10
Enclosure Front 10
Enclosure Mounting Tabs 11
USB Power and Communications 12
Internal Features 13
Battery 13
Internal Device Features 16
Enclosure Cover – LED Driver Board 18
Functionality 20
LED Indicators 20
Power 21
Battery 21
LTE 22
Push Notifications 23
Technical Specifications 25
Drawings 26

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Revision History

Prepared by	Date	Details of Revision	Revision
M. McMahon	04/12/2025	AC line power monitor prototype – User Manual.	0
M. McMahon	05/24/2025	Expanded instructions and warnings on handling of lithium polymer batteries. Update LED indicator functionality.	1

Approvals

Prepared by:

Mike McMahon – Cypress Embedded Systems

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Introduction

Welcome

Thank you for purchasing the PowerMonitor - AC Line Voltage Monitoring device by Cypress Embedded Systems.

This device is designed to monitor for the loss and restoration of AC line voltage and sending push notifications to the user via a cellular LTE network connection.

The device comes equipped with status indicators which provide status information at a quick glance.

Warranty Coverage

Cypress Embedded Systems ("Manufacturer") warrants that the **PowerMonitor** ("Product") will be free from defects in materials and workmanship under normal use for a period of **twelve (12) months** from the date of original purchase ("Warranty Period").

Exclusions:
This warranty **does not cover**:

i. The **supplied lithium battery**, which is provided "as-is" and without warranty.
ii. Damage resulting from misuse, abuse, neglect, accident, unauthorized modification, improper installation, or use outside the Product's intended purpose.
iii. Normal wear and tear or cosmetic damage.
iv. Damage caused by external factors such as power surges, water exposure, or extreme environmental conditions.

Remedy:
If a defect arises and a valid claim is received within the Warranty Period, Cypress Embedded Systems will, at its sole discretion, **repair or replace** the defective PowerMonitor or component with a new or refurbished part. Replacement parts or products assume the remaining warranty of the original Product or ninety (90) days, whichever is longer.

Limitation of Liability:
To the maximum extent permitted by applicable law, Cypress Embedded Systems' **total liability** under this warranty shall be **limited to the original purchase price** of the PowerMonitor. Under no circumstances shall Cypress Embedded Systems be liable for any indirect, incidental, consequential, or special damages, including but not limited to loss of profits, data, or use, arising from or related to the use or inability to use the PowerMonitor.

How to Obtain Warranty Service:
To make a warranty claim, contact Cypress Embedded Systems at mike@cypressembeddedsystems.ca or mike@3rdgear.ca with proof of purchase, a description of the issue, and your contact details. The PowerMonitor may need to be returned to the Manufacturer for evaluation. Please include the text "PowerMonitor Warranty Claim" in the subject line of your email when making a warranty claim.

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Governing Law:
This warranty shall be governed by and construed in accordance with the laws of the **Province of Ontario, Canada**.

Battery Liability Disclaimer

By using the **PowerMonitor**, which contains a **Lithium Polymer (LiPo) battery**, the user acknowledges and agrees to the following:

i. **Cypress Embedded Systems** ("Manufacturer") assumes **no liability** for any damage, injury, or loss arising from the use, handling, or malfunction of the **Lithium Polymer battery** inside the device.
ii. The **Lithium Polymer battery** is supplied **as-is**, and the Manufacturer makes no representations or warranties regarding its performance, safety, or suitability for any particular use.
iii. The user agrees to follow all manufacturer-recommended safety guidelines for the handling, charging, and disposal of the battery. The Manufacturer is not responsible for any consequences resulting from improper handling or failure to follow safety instructions.
iv. The Manufacturer is **not liable** for any direct, indirect, incidental, or consequential damages, including, but not limited to, injury, fire, or property damage, caused by the failure or misuse of the **Lithium Polymer battery**.

By using the **PowerMonitor**, the user assumes full responsibility for the care and safe use of the **Lithium Polymer battery** and agrees to hold **Cypress Embedded Systems** harmless from any liability related to the battery's performance or risks.

Cellular Connectivity and Return Policy

Please note that this device is **not returnable** due to its inability to connect to a LTE cellular network. It is the **customer's sole responsibility** to ensure that there is adequate cellular coverage in the area the PowerMonitor is intended to be installed **before** purchasing the device. Lack of connectivity resulting from insufficient network coverage, carrier compatibility, or other location-based limitations does **not** qualify as a valid reason for return or refund.

Please refer to the section LTE in this manual for additional information.

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General Considerations and Safety Information

Document Notation

This document uses four levels of notation to highlight important information and warnings.

DANGER	**Warns that failure to comply could result in serious injury or damage to property.**
WARNING	**Warns that failure to comply could result in minor or moderate injury or damage to property.**
CAUTION	**Warns that failure to comply may result in property damage.**
INFORMATION	**Notes provide additional information, emphasize a point, or provide a tip.**

Document Conventions

All dimensions referenced in this document are metric unless otherwise stated.

Operating Environment

The PowerMonitor device is intended for indoor use only. Use and store in a dry area at ambient room temperature (20-25° C), and out of direct sunlight.

Assumptions

It is assumed that users have the Pushover application installed on their smartphone and account set up. The Pushover application can be downloaded from the Apple App store (for iPhone) or Google Play (for Android). Installation and configuration of this application are outside the scope of this document. Users can access their Pushover account at www.pushover.net.

Cleaning

If the unit requires cleaning use a clothe *lightly* dampened with water only and wipe all exterior surfaces clean.

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WARNING

Do not use harsh chemicals, solvents, or detergents to clean this device.

Do not submerge this device in water or expose it to rain or adverse atmospheric conditions.

Mishandling from:

- **Dropping this device,**
- **Exposing this device to extreme temperatures or moisture,**
- **Using this device in a way that is not intended or specified in this manual, or**
- **Tampering with this device or attempting to repair it without authorization**

shall void all warranties implied or otherwise stated.

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Lithium Batteries

Many products we use today such as our smart phones, laptops, tablet computers, and other digital devices use lithium ion or lithium polymer batteries. While we may not think of it, these batteries come with risks with their use. Care must be taken when handling and using devices with these types of batteries installed.

Note that this product contains a lithium polymer battery.

DANGER

Do not drop, throw, modify, or tamper with lithium-ion batteries.

Do not use batteries that are swollen, dented, or show other signs of damage.

Store and use batteries in a dry location at room temperature, away from direct sunlight and extreme temperatures.

Dispose of batteries properly through designated recycling programs.

Do not overcharge batteries and ensure they are charged in a well-ventilated area, away from soft surfaces.

INFORMATION

This device charges and provides limited monitoring of the battery state as explained further in this manual.

DANGER

Do not use any other means of charging the internal lithium battery. Using an external charger can result in damage to the battery, the PowerMonitor device, and/or personal property.

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Periodic Checks

The internal lithium polymer battery should be inspected every three (3) months. A visual inspection should look for:

- Swollen, dented battery.
- Any signs of physical damage.

INFORMATION

Nominal battery voltage is 3.7Vdc. A fully charged battery should be approximately 4.2Vdc.

CAUTION

If the battery has been charged for a long period of time and the battery is not holding a charge, or the voltage is less than 3.2Vdc, replace the battery.

DANGER

Do not continue to use lithium ion/lithium polymer batteries that show signs of damage or will not hold a charge.

If required, batteries can be purchased from Cypress Embedded Systems. Contact mike@cypressembeddedsystems.ca. A replacement battery can also be purchased directly from https://ca.robotshop.com/.

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Device Features

External Features

The PowerMonitor is housed in an enclosure with dimensions as shown in the section Drawings. External features of the device are shown in the figures below.

Enclosure Front

Figure 1 - Enclosure Front

1- (Qty 4) M3 x 8mm Flat Head Screws (hex socket)
2- Power Status indicator LED
3- Battery Status indicator LED
4- Cellular (LTE) Status indicator LED

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Enclosure Mounting Tabs

Figure 2 - Mounting Tabs

The PowerMonitor includes four mounting tabs; two at the top (not shown) and two at the bottom. The holes will accept screws up to a diameter of 5mm.

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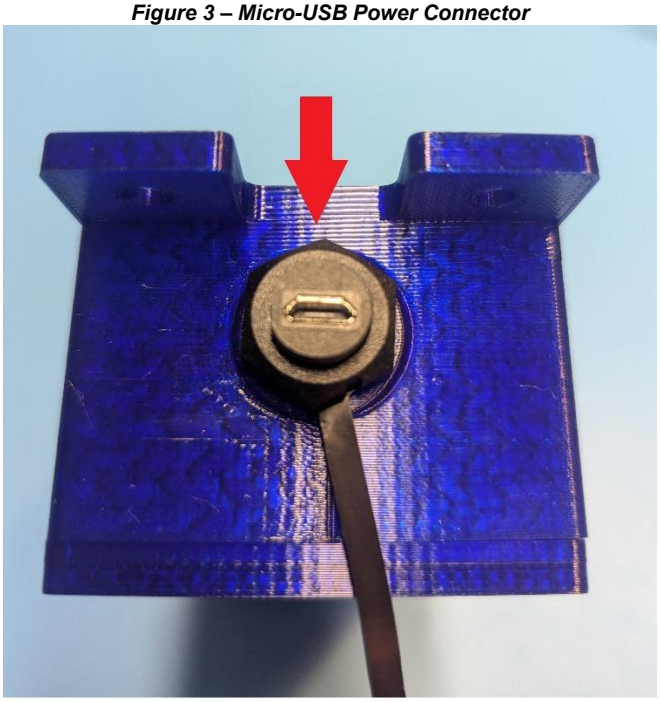
USB Power and Communications

Figure 3 – Micro-USB Power Connector



The PowerMonitor includes a Micro-USB connector at the top (with dust cover). The connector's primary use is to power the device with the included DC power adaptor. Additionally, the Micro-USB connector allows the device to connect to a computer for firmware updates.

INFORMATION

The included DC power adaptor is rated for 5Vdc, 3A and has a male Micro-USB connector that plugs into the PowerMonitor's power connector. If a replacement power adaptor is needed, ensure its output is 5Vdc and at minimum 1A.

CAUTION

Ensure the supplied DC power adaptor is kept in good condition and free from excessive wear or damage. Replace the power adaptor if it has been mishandled or shows signs of wear or damage.

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Internal Features

The following describes the internal features of the device.

Battery

Figure 4 - Lithium Polymer Battery

The PowerMonitor is supplied with a single cell, 3.7Vdc, 1800mAh lithium polymer battery. The battery serves as a backup power supply for the PowerMonitor when AC line voltage is lost. The supplied battery can provide approximately 24 hours of backup power.

INFORMATION

The battery leads and connector are polarized. The positive (**+**) wire is **RED**, and the negative (**-**) wire is **BLACK**. This is important to note when connecting the battery to the main control board inside the PowerMonitor enclosure.

DANGER

Do not continue to use lithium ion/lithium polymer batteries that show signs of damage or will not hold a charge.

DANGER

Do not drop, throw, modify, or tamper with lithium-ion batteries.

Do not use batteries that are swollen, dented, or show other signs of damage.

Store and use batteries in a dry location at room temperature, away from direct sunlight and extreme temperatures.

Dispose of batteries properly through designated recycling programs.

Do not overcharge batteries and ensure they are charged in a well-ventilated area, away from soft surfaces.

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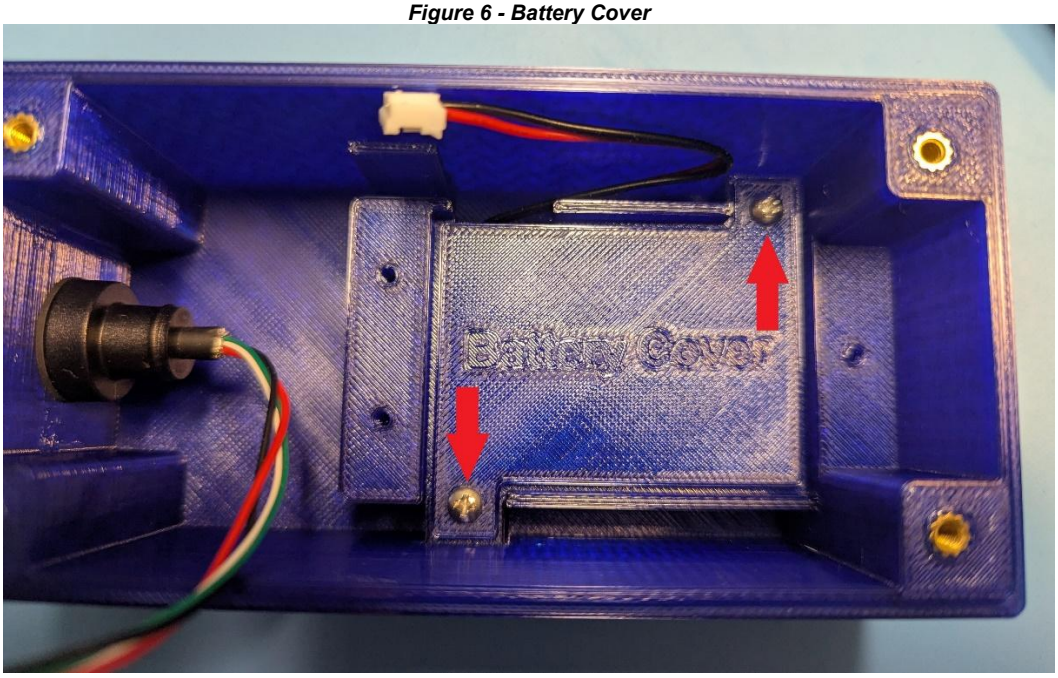
Battery Installation

Figure 5 - Battery Compartment

The battery is located inside the enclosure as shown in Figure 5 - Battery Compartment.

Figure 6 - Battery Cover



The battery cover as shown in Figure 6 - Battery Cover is secured using two M3 x 10mm 18-8 stainless steel thread forming screws for plastic.

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CAUTION

If the M3 x 10mm 18-8 stainless steel thread forming screws for plastic are removed care must be taken when reinstalling them to ensure that new threads are not cut into the plastic body of the enclosure. Failure to do this will result in damage to the enclosure which is not covered by warranty.

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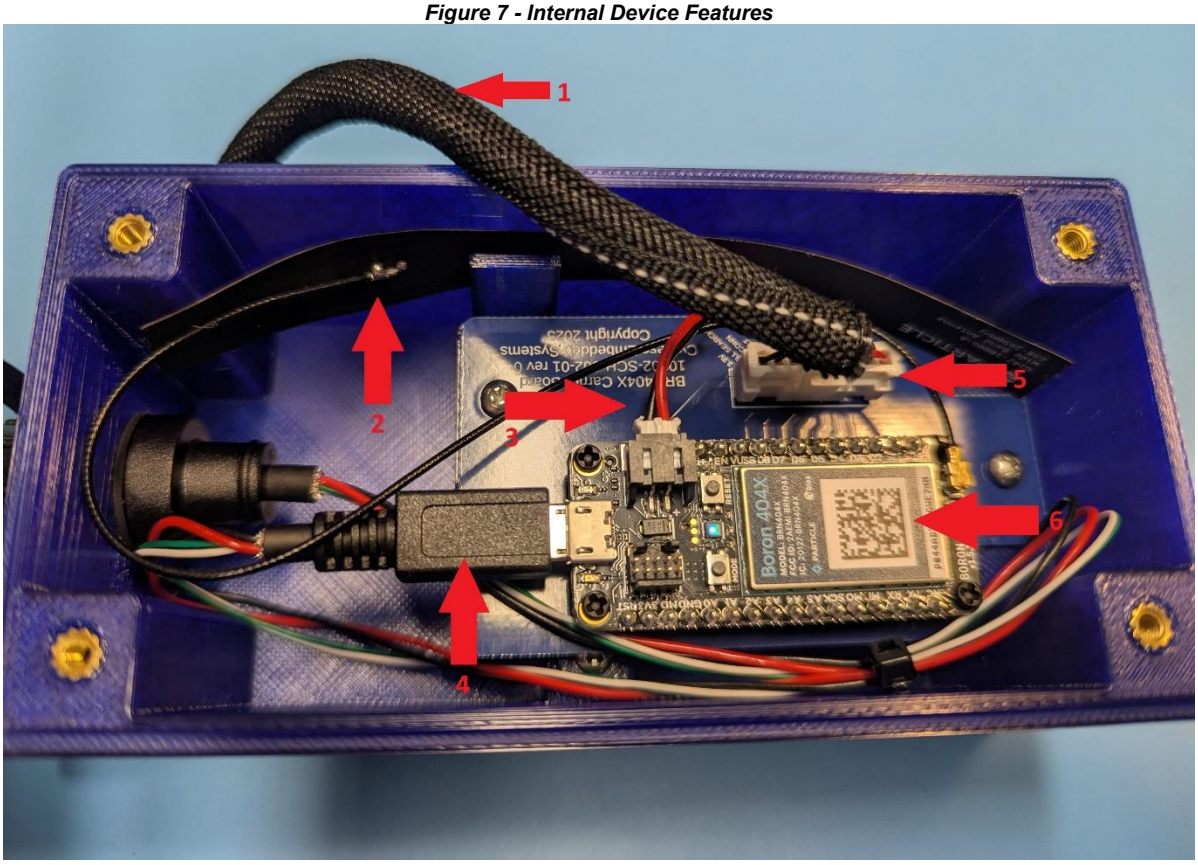
Internal Device Features

Figure 7 - Internal Device Features



The internal features of the PowerMonitor device are as follows.

1 – LED indicator board wiring harness
2 – LTE cellular antenna
3 – Lithium polymer battery connector
4 – Main control board Micro-USB connector
5 – LED indicator wiring harness connector and header (to main control board)
6 – Particle Boron BRN404X main control board

The LTE cellular antenna has an adhesive backing which has not been used in order to facilitate easier access to the battery compartment. The enclosure has been designed with a retaining clip to hold the antenna against the enclosure side wall.

INFORMATION

The lithium polymer battery JST connector is polarized. The **RED** wire is positive (**+**), and the **BLACK** wire is negative (**-**).

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Figure 8 - 2-Pole JST Battery Connector

INFORMATION

The 2-pole JST connector is polarized and features a tab at the top of the housing that allows the connector to insert into the main control board only one way. This tab must be pointed upward.

CAUTION

IMPORTANT:
The JST connector is polarized and only inserts in one way onto the main control board.

Note the orientation of the device body and of the JST connector shown in Figure 7 - Internal Device Features and ensure the RED positive (+) lead is on the right, and the BLACK negative (-) lead is on the left.

Not all batteries come with the connector attached in the correct polarity.

Failure to observe correct connector polarity will result in permanent damage to the main control board and to the battery.

The main control board is secured to the enclosure using three M3 x 10mm 18-8 stainless steel thread forming screws for plastic.

CAUTION

If the M3 x 10mm 18-8 stainless steel thread forming screws for plastic are removed care must be taken when reinstalling them to ensure that new threads are not cut into the plastic body of the enclosure. Failure to do this will result in damage to the enclosure which is not covered by warranty.

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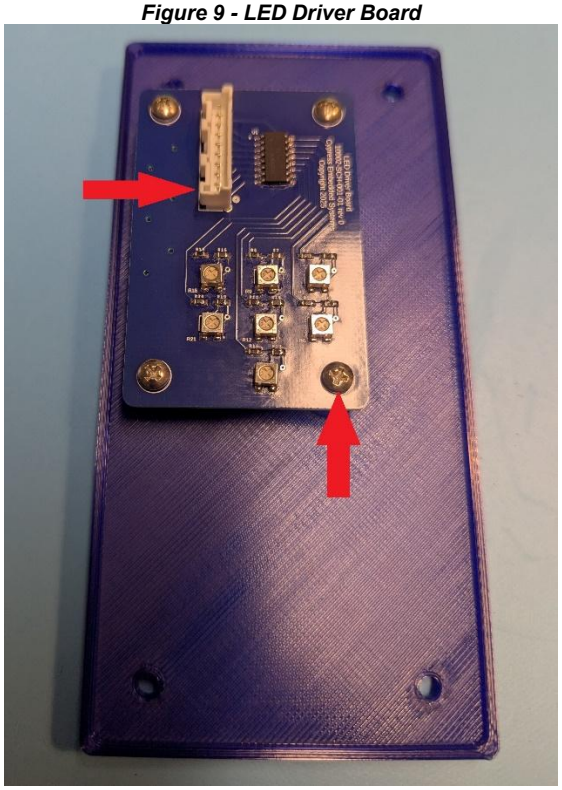
Enclosure Cover – LED Driver Board

Figure 9 - LED Driver Board



The PowerMonitor cover holds the LED driver circuit that powers the LED indicators on the front cover.

The header on the LED driver circuit board connects to the opposite end of the wiring harness shown in Figure 7 - Internal Device Features.

The LED driver circuit board is secured to the cover using four M3 x 10mm 18-8 stainless steel thread forming screws for plastic.

CAUTION

If the M3 x 10mm 18-8 stainless steel thread forming screws for plastic are removed care must be taken when reinstalling them to ensure that new threads are not cut into the plastic body of the enclosure. Failure to do this will result in damage to the enclosure which is not covered by warranty.

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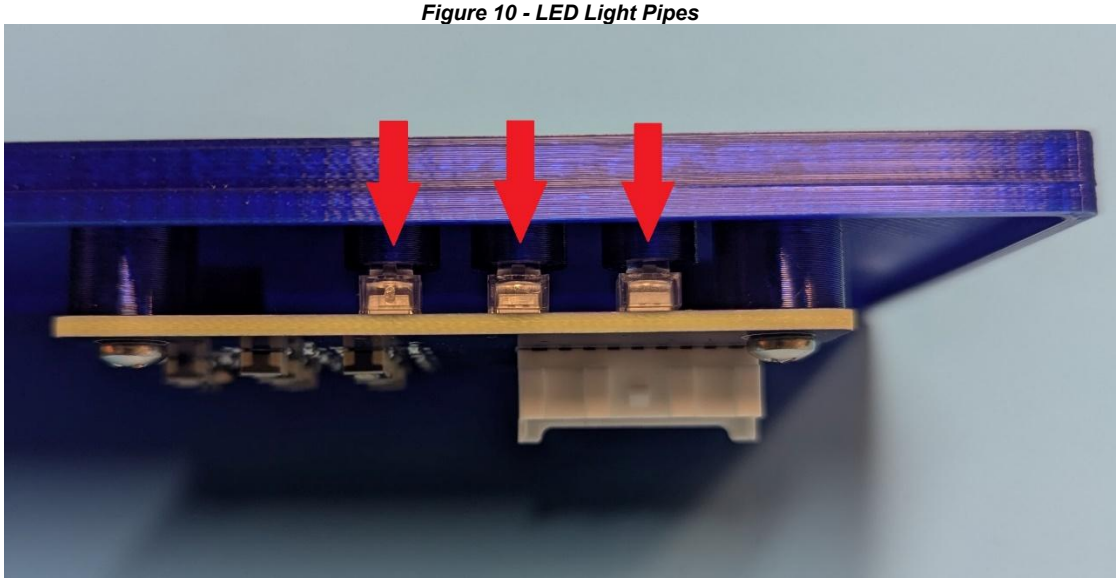
INFORMATION

NOTE: The LED's used are of the surface mounted type. LED light pipes are used to project the light from the LEDs to the front of the enclosure cover. These light pipes are loose fit and will fall off if the LED driver circuit board is removed from the cover. When reinstalling the light pipes note that there are bosses on them that match to mounting holes on the circuit board. See Figure 10 - LED Light Pipes.

Figure 10 - LED Light Pipes



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Functionality

The PowerMonitor is designed to provide users push notifications to their smartphones when AC line voltage is lost or restored.

INFORMATION

For the device to function properly, make sure the device DC power adaptor is plugged into a 120Vac power outlet.

Ensure that the power outlet is not protected with a battery backup system such as an Uninterruptible Power Supply (UPS).

LED Indicators

The LED indicators provide users with valuable status information about the condition of the PowerMonitor device.

Table 1 - LED State Colors

LED	State		
Power			
Battery			
LTE			

Table 2 - LED State Color Legend

Indicator Legend	
	Solid Green
	Flashing Yellow (0.5Hz)
	Solid Red
	Flashing Red (0.5Hz)

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The following describes the meaning of each LED indicator.

Power

When the PowerMonitor is operating on AC line voltage this LED will be solid green in color. When AC line voltage is lost the LED indicator will turn solid red.

Battery

When the internal lithium polymer battery is fully charged the LED indicator will be solid green in color. If the battery is charging, the LED indicator will flash at a frequency of 0.5Hz and will be yellow in color. If there is a problem with the battery the LED indicator will be solid red in color.

Note that when the PowerMonitor is running from the backup battery, the Battery LED will not be on unless there is a battery fault.

If the Battery LED is red, this indicates that there is a problem with the battery. The following are conditions that can cause a battery fault.

Battery State Unknown
Battery Not Charging
Battery Fault
Battery Not Connected

If a fault exists, check to ensure that the battery is connected to the main control board. Alternatively, if the PowerMonitor has been in service for a long period of time, open the battery cover and inspect the battery. If the battery is puffy or appears deformed, not holding a charge (low voltage after being charged for a long time), disconnect the battery and dispose of it in an appropriate manner. Replace the battery.

DANGER

Do not continue to use lithium ion/lithium polymer batteries that show signs of damage or will not hold a charge.

Replacement batteries can be purchased from Cypress Embedded Systems. Contact mike@cypressembeddedsystems.ca. Alternatively, a replacement battery can be purchased directly from https://ca.robotshop.com/.

INFORMATION

Nominal battery voltage is 3.7Vdc. A fully charged battery should be approximately 4.2Vdc.

The normal operating range of the battery is 3.2Vdc minimum to 4.2Vdc maximum.

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LTE

When the PowerMonitor is connected to both cellular LTE and the Particle.IO cloud service, the LED will be solid green. When the device first powers on, the LED will be solid red until cellular LTE and Particle.IO cloud service is connected.

While the PowerMonitor is running it checks to verify that it is still connected to both cellular LTE and the Particle.IO cloud service every 60 seconds. If either service is lost, the LTE LED will flash red at a frequency of 0.5Hz until these services are reconnected.

INFORMATION

No push notifications will be received if there is no connection to cellular LTE and Particle.IO cloud services.

If you are unable to connect to the cellular network, try moving the PowerMonitor to a different location. This device will work in most basements; however, try to keep the device out of areas where a cellular network signal may become attenuated such as areas with lots of concrete.

INFORMATION

The following links provide cellular coverage maps for Bell Canada, Telus Mobility, and Rogers. The PowerMonitor uses LTE Cat M1 for IoT devices.

https://www.bell.ca/Mobility/Our_network_coverage

https://www.rogers.com/mobility/network-coverage-map

https://www.telus.com/en/mobility/network/coverage-map

INFORMATION

The area and environment where your PowerMonitor is installed influences cellular signal strength and quality.

If you have installed the PowerMonitor in an area that has a weak cellular signal you may show being connected to LTE on the device, but data may not get transmitted. Push notifications may not be received. If your cellular phone shows a weak signal this can possibly confirm this scenario.

If moving the device to another area is unsuccessful, consider installing a cellular signal booster. These are available at various online retailers such as Amazon. When selecting one, ensure it has support for Bell, Rogers, and Telus, and for cellular LTE bands 2, 4, 5, 12, and 13.

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Push Notifications

Users of the PowerMonitor are required to have the Pushover application installed and running on their smartphone. Installation and configuration of the Pushover application are outside the scope of this document.

The following are the push notifications that the PowerMonitor provides.

Initial Power Up:

Application booting… AC POWER GOOD
Application booting… NO AC POWER

When the PowerMonitor is first powered on, one of the above push notifications will be received.

Normal Operation:

Power Monitor – AC POWER LOST
Power Monitor – AC POWER IS ON

Under normal operating conditions, one of the above push notifications will be received upon loss or restoration of AC line voltage.

Battery Status – Fault Conditions:

Battery State Unknown
Battery Not Charging
Battery Fault
Battery Disconnected

If there is an issue with the internal lithium polymer backup battery, one of the above push notifications may be received. Battery issues should be addressed at the earliest possible opportunity.

Low Battery Power:

Power Monitor – Low Battery
Power Monitor – Shutting Down…
Power reconnected – Resuming operation

INFORMATION

Push notifications are date and time stamped.

The PowerMonitor device measures the internal battery voltage while it is operating on battery backup. If the battery's voltage drops below a minimum threshold a push notification will is sent to the user warning that such a condition exists. Shortly after this message is sent the PowerMonitor device will shut itself down to protect the unit from an undervoltage condition that can permanently damage the battery.

If AC line voltage is restored while in the process of shutting down, the shutdown procedure will halt and normal operation will resume; otherwise, the unit shuts down and will only power back on when AC line voltage is restored.

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The PowerMonitor device handles the charging of the internal lithium battery while AC line voltage is present. The device will charge the battery and maintain a charged state. Internal power management circuitry prevents the battery from overcharging.

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Technical Specifications

Enclosure	NEMA Type 1 / IP10 equivalent custom 3D printed enclosure (Spectrum Filaments FlameGuard ASA 275).
PowerMonitor Input Power	Input power via microUSB connector; 5Vdc, 500mA maximum.
DC Power Supply	Input: 120Vac Output: 5Vdc, 3000mA
Cellular	Embedded EtherSIM for LTE Cat M1 North America (bands 2, 4, 5, 12, 13) **Device to use supplied antenna only.** **Device is certified for use with the supplied antenna only.**
Particle Boron BRN404X	**Canada Statement:** This device complies with Industry Canada's licence-exempt RSSs. Operation is subject to the following two conditions: 1. This device may not cause interference; and 2. This device must accept any interference, including interference that may cause undesired operation of the device. Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: 1. l'appareil ne doit pas produire de brouillage; 2. l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement. Caution Exposure: This device meets the exemption from the routine evaluation limits in section 2.5 of RSS102 and users can obtain Canadian information on RF exposure and compliance. Le dispositif répond à l'exemption des limites d'évaluation de routine dans la section 2.5 de RSS102 et les utilisateurs peuvent obtenir des renseignements canadiens sur l'exposition aux RF et le respect. **The final end product must be labelled in a visible area with the following:** The Industry Canada certification label of a module shall be clearly visible at all times when installed in the host device, otherwise the host device must be labelled to display the Industry Canada certification number of the module, preceded by the words "Contains transmitter module", or the word "Contains", or similar wording expressing the same meaning, as follows: **Contains transmitter module ISED: 20127-BRN404X** This End equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body. Cet équipement devrait être installé et actionné avec une distance minimum de 20 centimètres entre le radiateur et votre corps. The end user manual shall include all required regulatory information/warning as shown in this manual. **Refer to the following website for further details about this module:** https://docs.particle.io/reference/datasheets/b-series/brn404x-datasheet/

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Drawings

The following pages include dimensional drawings for:

i. Enclosure body
ii. Enclosure lid
iii. Battery compartment cover
iv. LED driver board
v. Main control board

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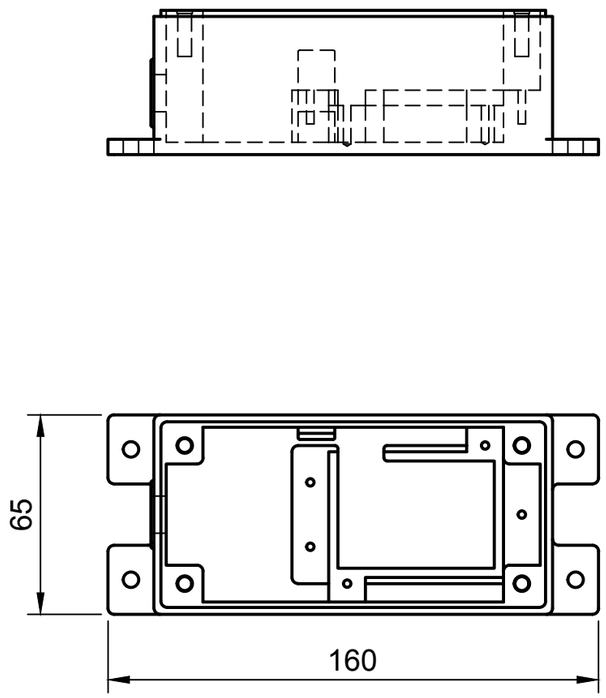
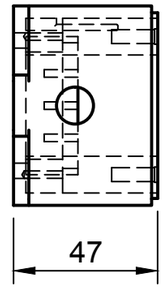
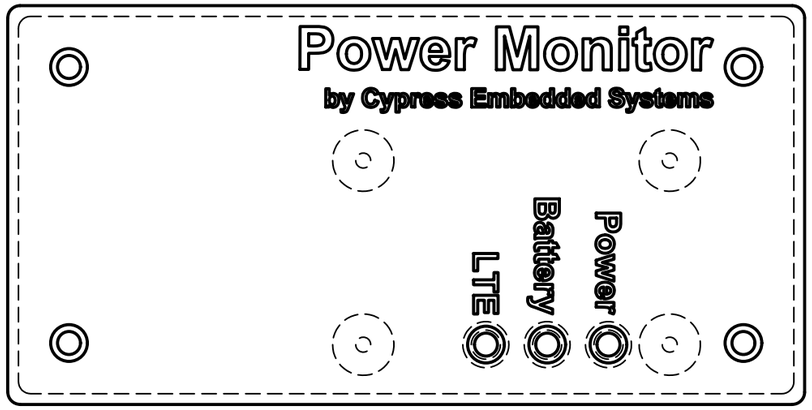
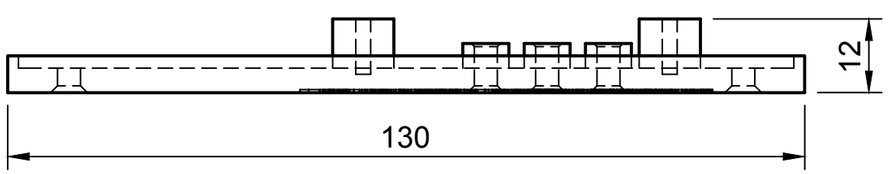
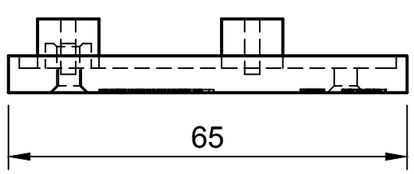




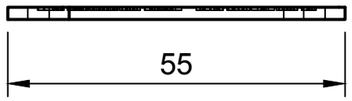
Dept.	Technical reference	Created by Mike Mcmahon 2025-04-06	Approved by	
		Document type	Document status	
		Title Power Monitor Enclosure Bottom	DWG No.	
			Rev. / Date of issue	Sheet 1/1

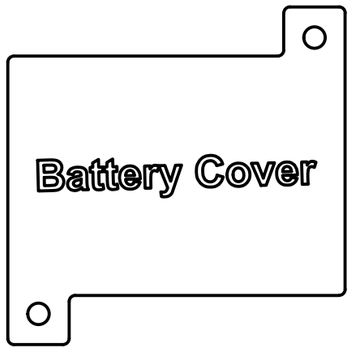


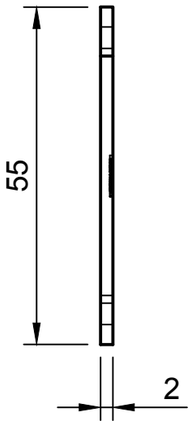




Dept.	Technical reference	Created by Mike Mcmahon 2025-04-06	Approved by
		Document type	Document status
		Title Enclosure Lid	DWG No.
			Rev. / Date of issue / Sheet 1/1







Dept.	Technical reference	Created by Mike Mcmahon 2025-04-06	Approved by	
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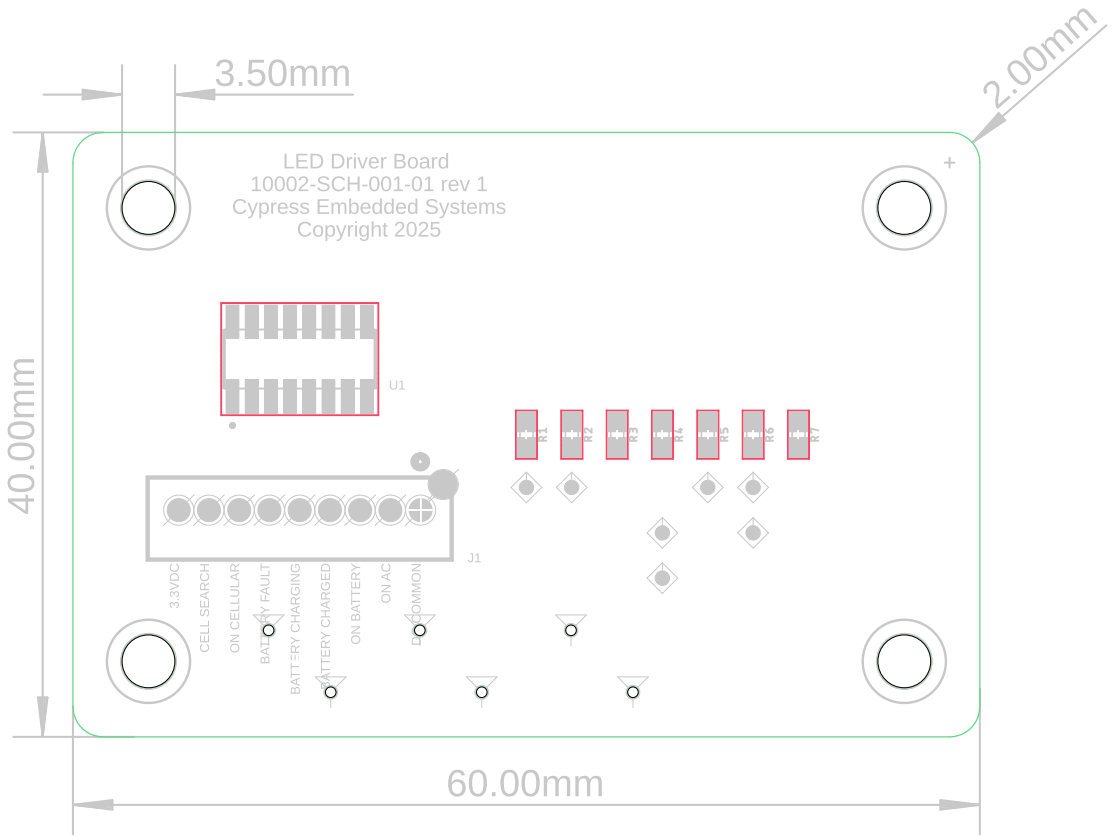
		Title Battery Cover	DWG No.	
			Rev. / Date of issue	Sheet 1/1


3.50mm
2.00mm
40.00mm
60.00mm
LED Driver Board
10002-SCH-001-01 rev 1
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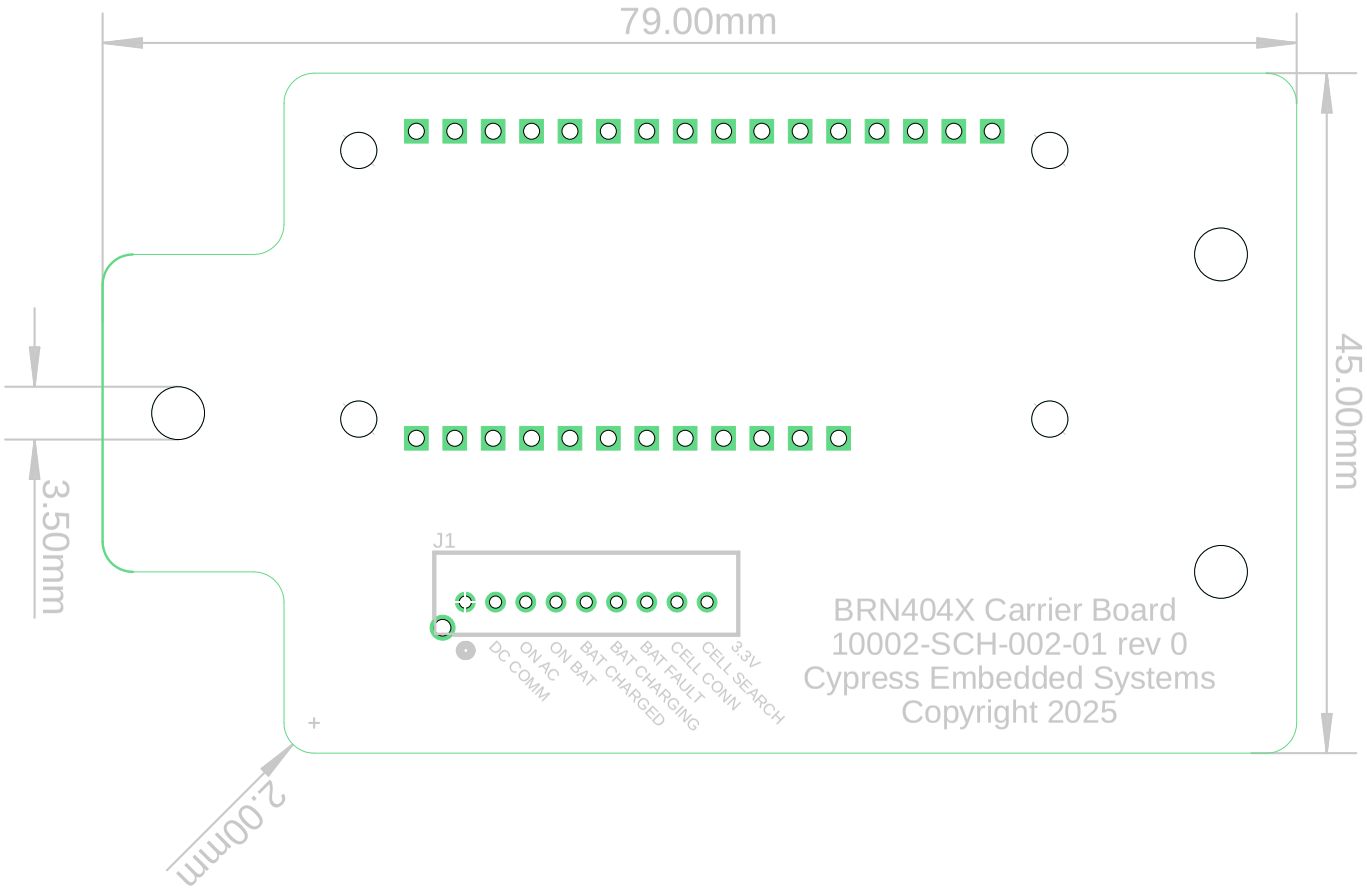
U1
J1
R1
R2
R3
R4
R5
R6
R7
3.3VDC
CELL SEARCH
ON CELLULAR
BATTERY FAULT
BATTERY CHARGING
BATTERY CHARGED
ON BATTERY
ON AC
LED COMMON


79.00mm
45.00mm
3.50mm
2.00mm
J1
DC COMM
ON AC
ON BAT
BAT CHARGED
BAT CHARGING
BAT FAULT
CELL CONN
CELL SEARCH
3.3V
BRN404X Carrier Board
10002-SCH-002-01 rev 0
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